Diffusion Processes for Inference of Structural Causal Models

IA pour la Science

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1 Linear models excluding State-Space models

List of SCM-model parameters:

w the linear parameters of the model,

v the non-linear parameters of the model,

the covariance (mutual information) of the variables $X_1, \ldots, X_n$,

1. the covariance of the linear part of the parameters **w**,
2. the covariance of the non-linear part of the parameters **v**.

The transformations

1. only G to f it admissible (there is no conversion from the model to G)

Architectures of the linear models:

1. linear model $\hat{\mathbf{x}} = \mathbf{W}^\mathsf{T}\mathbf{x}$: a single matrix (2-index) that maps $X_1, \ldots, X_n$ to $\hat{X}_1, \ldots, \hat{X}_n$,
2. generalized linear model $\boldsymbol{\sigma} \circ \mathbf{W}^\mathsf{T}\mathbf{x}$: non-parametric link function of the linear model,
3. special non-linear model $g_1(\mathbf{v}_i) \circ g_r(\mathbf{v}_r)(\mathbf{x})$: a model with functions $g_i(\mathbf{v}_i)(X_j)$,
4. general non-linear model $\mathbf{W}^\mathsf{T} g_i(\mathbf{v}_i)(X_j)$ (obsoleted): a linear combination of the model with functions,
5. generalized non-linear model $\boldsymbol{\sigma} \circ \mathbf{W}^\mathsf{T} g_i(\mathbf{v}_i)(X_j)$: a non-linear non-parametric transformation of the linear combination

Optimization:

1. for linear models is MLS,

2. for non-linear models is Levenberg-Marquardt,
3. for the non-linear models is Levenberg-Marquardt,
4. for the rest it will

Here the notation $\circ$ means the composition of functions. It could be replaced by brackets.